Final Technical Report:
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Link to Github

Introduction:

Our group decided to research if weather has an effect on energy usage in a household. With an accurate model, we would be able to account for houses more affected by weather and find solutions to reduce the amount of electricity from these conditions. We found a dataset from a house in Northern Mexico that gave us the ability to test our thoughts. This dataset took the energy usage from that house every minute for 14 months straight; over 612,000 data points, giving us plenty of data to work with.

Data:

While looking for good data for energy usage and environmental factors, we found a good article on Science Direct that studied residential energy consumption and environmental factors. That article had a link to the dataset, which is where we were able to see the data and what the variables mean, which is listed below.
The variables in the data set are:

- date (The day and hour the data was taken)
- active_power (Active Power)
- current (The electrical current)
- voltage (The Electric tension)
- reactive_power (Computed reactive power Q=SQRT(S^2-P^2)
- apparent_power (Computed Apparent Power S=V*I)
- power_factor (Computed Power Factor PF=P/S
- main (General description of weather)
- description (Detailed description of the Weather)
- temp (Temperature)
- feels_like (Temperature sensation)
- temp_min (Minimum temperature)
- temp_max (Maximum temperature)
- pressure (Air pressure)
- humidity (Amount of moisture in the air)
- speed (Wind Speed)
- deg (Wind Direction)
- temp_t+1 (Forecasted temperature)
- feels_like_t+1 (Forecasted Temperature sensation)

Our target variable is Active power, and the independent variables are the rest listed above.

Preprocessing and Exploratory Data Analysis:

Before modeling, we realized that we cannot have the date variable in the model because it would predict using the minute after or the minute before, resulting in a model that was inaccurate and overfit.

The next step was to convert all temperature variables from Celsius to Fahrenheit, and drop 'temp_t+1', 'feels_like_t+1', because they are not needed for analysis. After that, we one-hot encoded the 'main' variable because it is categorical and had a few outcomes. This allows the model to use these weather patterns to help with its predictions.

We then dropped any variable dealing with the electricity, as it would overfit the model and ruin the model. These had to be dropped because the model could just create a formula from these variables to calculate the exact amount of power the house was using. ('power_factor', 'apparent_power', 'reactive_power', 'voltage', 'current')

Modeling:

To create a testing dataset, first, we had to set X to all the variables from the preprocessed data without current power and set y to the target variable, which was current power. After this was done, we can do a train test split for the data, which creates a 20% testing set and an 80% training set. This step has to be done so the model has something to predict and doesn't overfit the model. Afterwards, we made 3 different models using 3 different copies of the training set and testing sets. The 3 models we created were a Linear Regression model, A Decision Tree Regressor model, and a Tuned Decision Tree.

Results:

For the first model, we chose Linear Regression, as we have not talked about it too much in class, and we wanted to test it out. This model finds a line that minimizes the sum of squared differences between the predicted and actual values, along with the coefficient for each variable to represent how much they affect active power. After running the model, we obtained an R^2 score of 0.123. And an MSE of 31505.093. An r^2 value shows low accuracy and shows the model or data set needs work. We made a barplot of the coefficients on the accuracy of the model, as seen in Figures 1 and 2. Then calculated the residuals for the residual plot.

For the second model, we made a Decision Tree Regressor. Decision trees work by creating splits in order to minimize variance in the resulting groups, which form the model's predictions. After running the Decision Tree Regressor, we came out with an r^2 score of 0.607 and an MSE of 14112.515. An r^2 value of .6 shows this model is significantly better than the last model. (See figures 3,4 and 5). After we made a barplot to show the importance of each variable and how much weight it carries in the decision tree. And a Residual Plot after, just like the model before.

The third model was a Tuned Decision Tree. We set different max depths and min leaf samples. We resulted in an r^2 score of 0.167 and an MSE of 29946.393. Once again, the r^2 value is low. This tuned tree is much simpler but results in worse performance, most likely due to the second model possibly being overfit. (See figures 6,7,8)

Conclusion:

From the results of our research, we can conclude that it is very difficult to predict the amount of electricity a house is going to use based on the weather it is currently facing. This is likely due to the uncertainty of the weather and the difficulty of trying to predict a value based on another varying factor. An interesting part of the model was that the most important factor in our best model was using feels like temperature to predict the energy usage. Since this is what the weather outside is actually feeling like, compared to numbers that have a less impactful value, which makes a lot of sense. With the second most important factor being wind direction, it is possible that this is something that only applies to that specific house in Northern Mexico. It is challenging to know if a finding can be generalized when all the data is coming from one house in Northern Mexico. However, our findings from this project can still be used, even if it's finding a solution for houses in that neighborhood to save money on their electrical bill based on the weather they are experiencing.

Limitations and Future Work:

While there were not many problems throughout the project, the biggest problem came from the unpredictability of the weather in general. It is very hard to predict another numerical value based on another factor with a lot of uncertainty. This made creating accurate predictions and models very difficult.

In conclusion, our best model, the Decision Tree Regressor, was moderately accurate and could give an estimate of the power output.

Our main limitation throughout this project was that our data is specific to a house in Northern Mexico, which can heavily limit the global impact of this project. It would also make our results not comparable to other regions, whether the results were good or bad.

Transparency:

Claude Sonnet 4.5 was used for small debugging problems during the coding portion of this project.

Code:

GitHub Repository

References:

- Dataset: Mendeley Data
- Source Article: ScienceDirect

Python Libraries Used:

- pandas
- numpy
- matplotlib
- seaborn
- scikit-learn

Figure 1

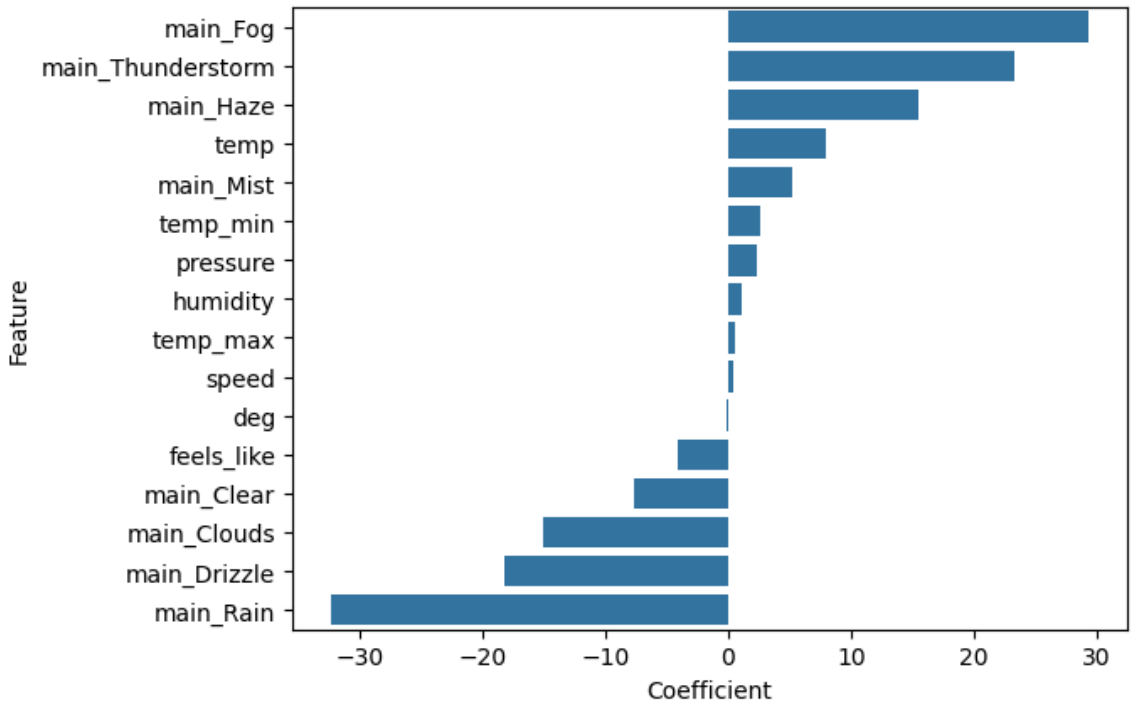

Figure 2

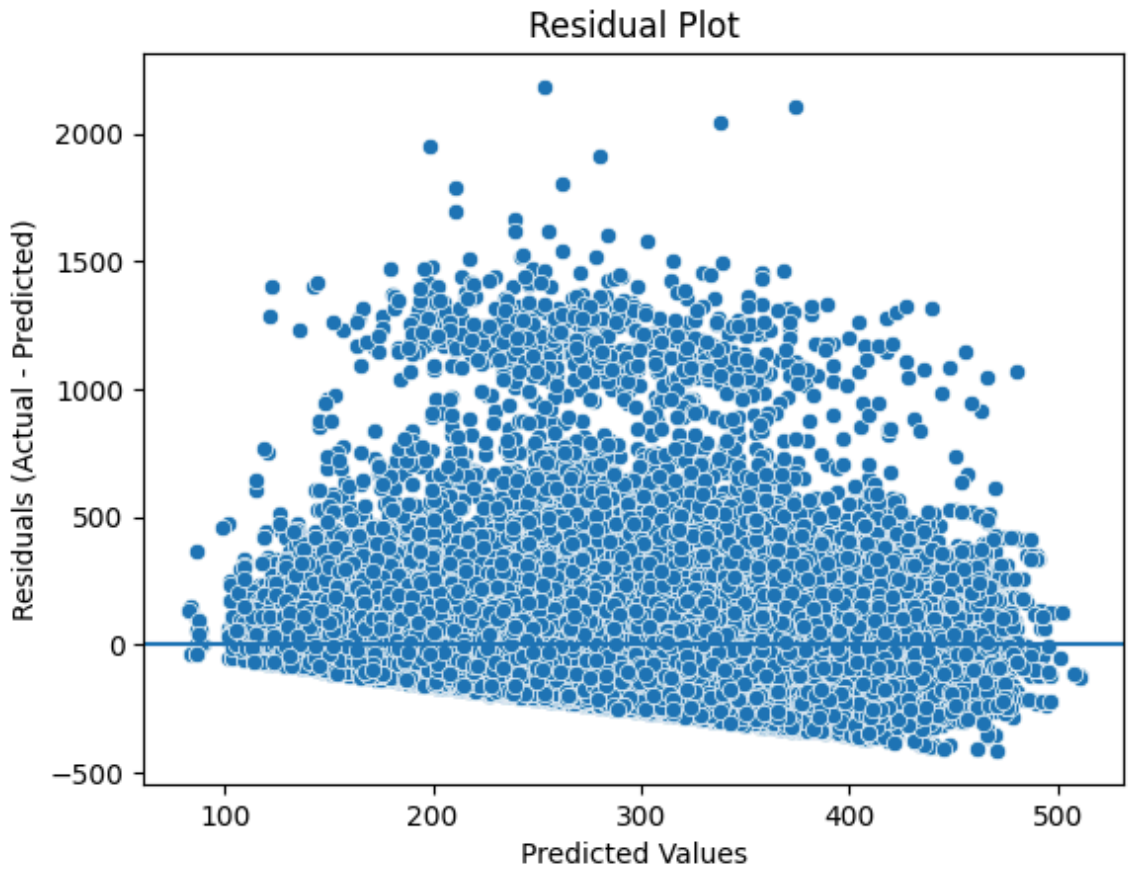
Residual Plot
Residuals (Actual - Predicted)
2000
1500
1000
500
0
−500
100
200
300
400
500
Predicted Values

Figure 3

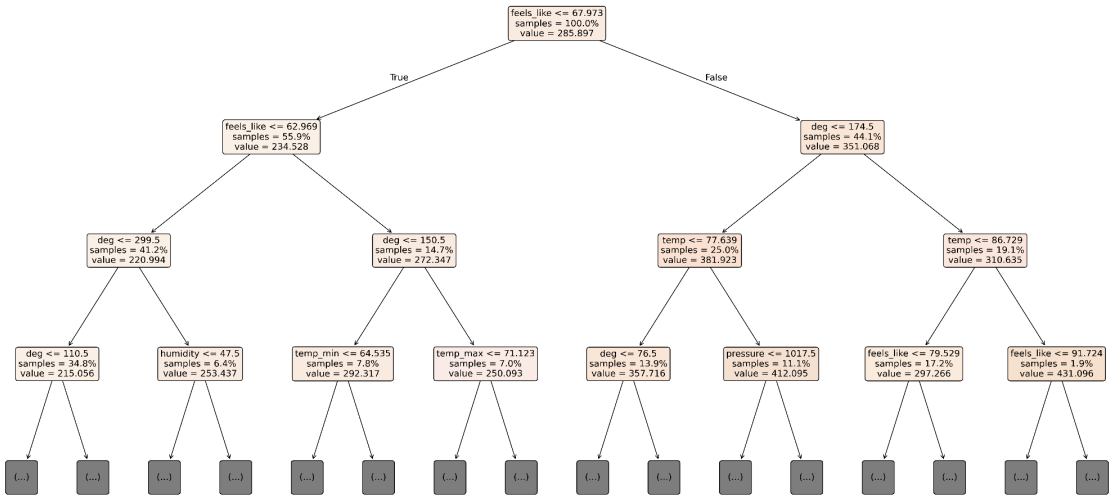

Figure 4

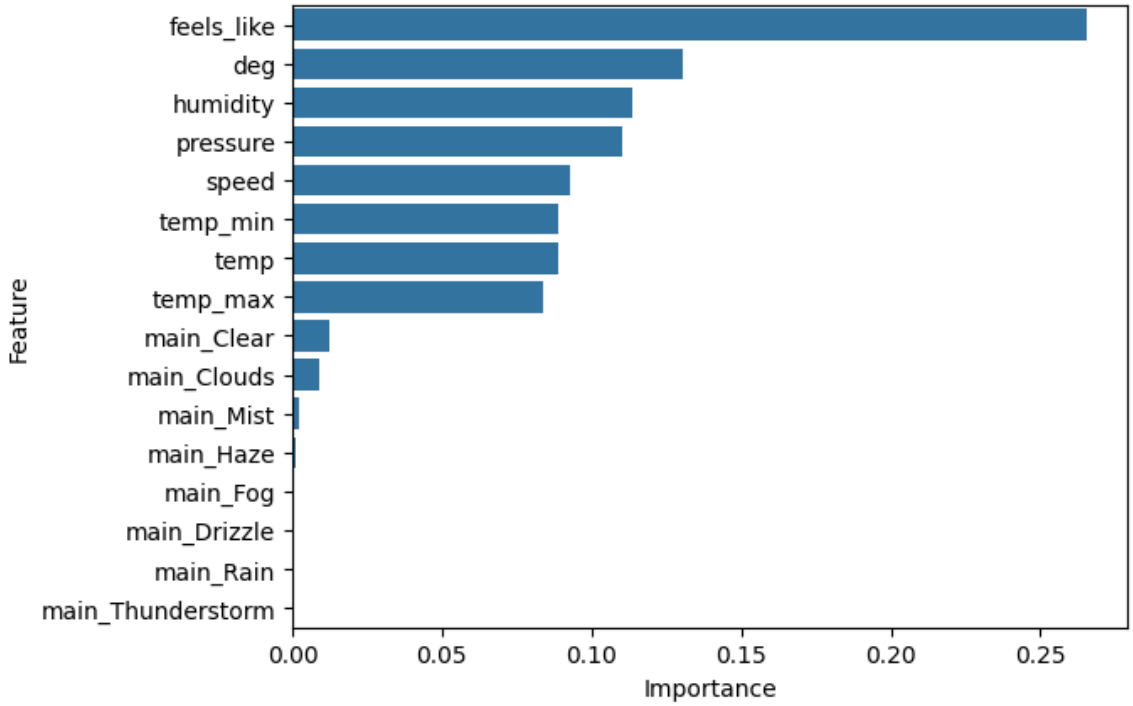

Figure 5

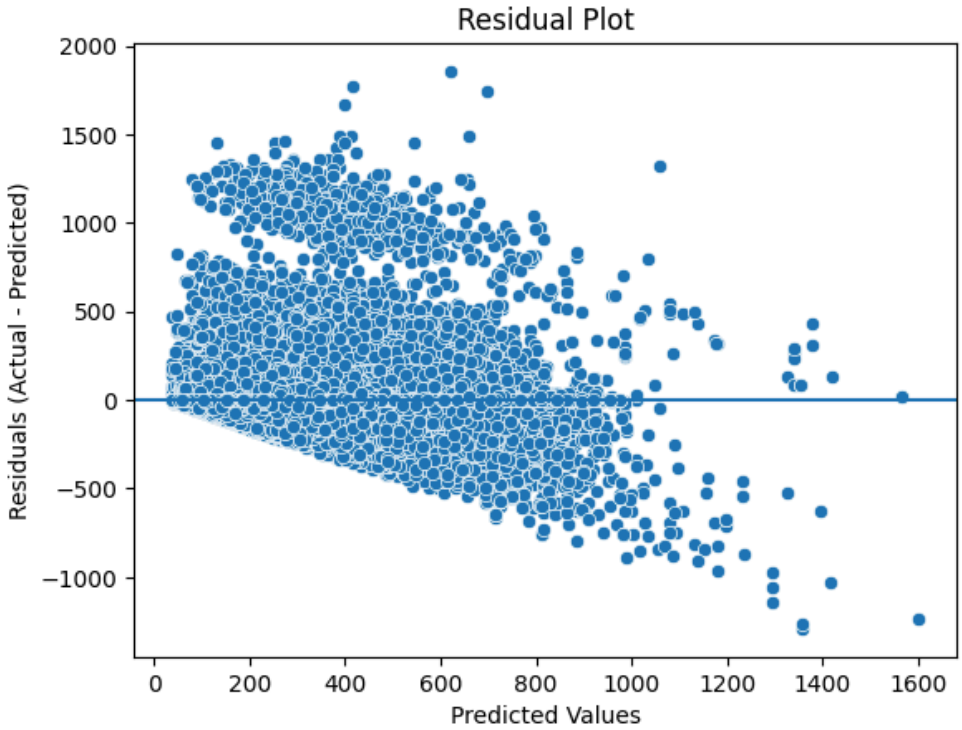

Figure 6

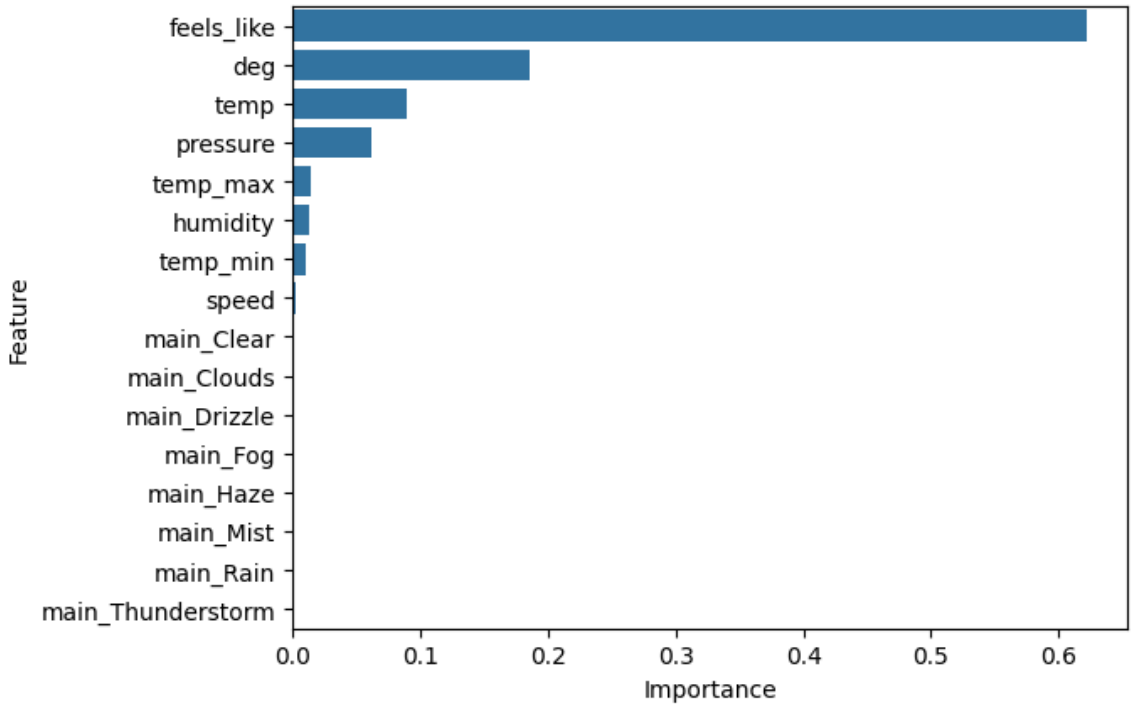

Figure 7

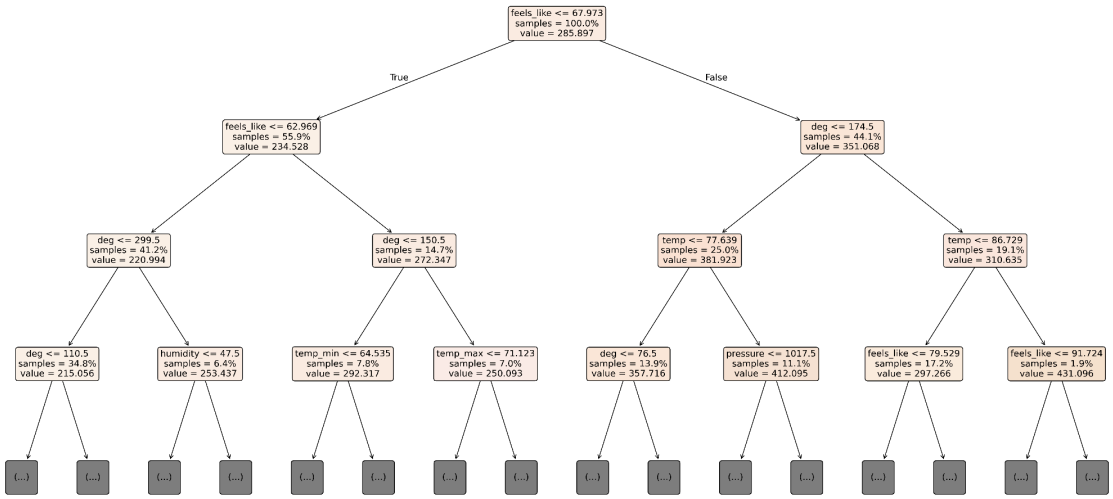
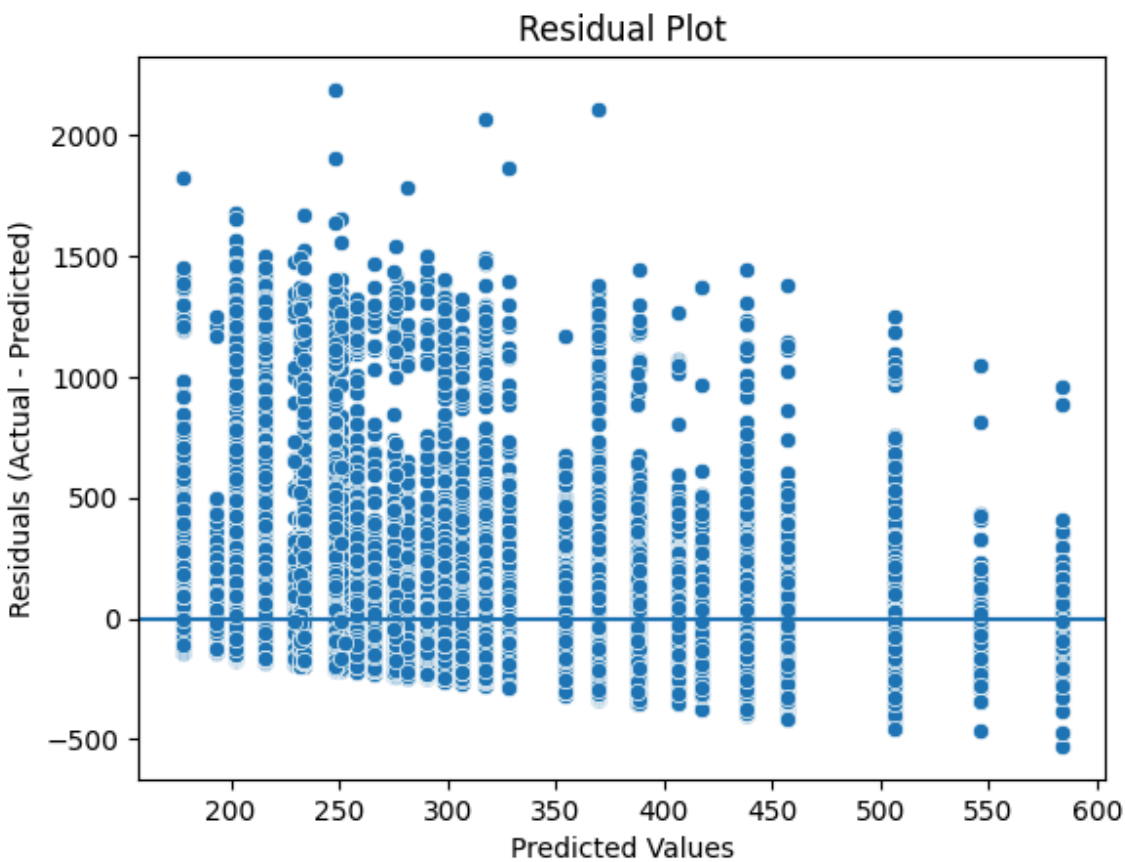
Figure 8